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Department of Health Research
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## Standard Treatment Workflow (STW)

# RESPIRATORY DISTRESS IN NEONATES

**ICD-11: KB23**

### RESPIRATORY DISTRESS

Presence of ANY ONE:

- RR ≥60/min
- Chest retractions
- Nasal flaring
- Grunting

### IMMEDIATE ACTIONS

**Apnoea/gasping, poor effort or shock: Stabilize immediately; ventilate when indicated**

- Maintain TABC; provide warmth; admit to SNCU/NICU
- Attach pulse oximeter; monitor HR, RR, $SpO_2$ and CRT; assess SAS
- Feed when stable: breastfeed if mild RD; OG feeds when appropriate on CPAP; decompress stomach
- Withhold feeds; give IV fluids for severe RD, recurrent apnoea, poor perfusion or abdominal concerns

### ALGORITHM

**ASSESS GESTATIONAL AGE (GA)* + SILVERMAN–ANDERSEN SCORE (SAS)**

*If GA is uncertain, BW ≤1800 g may be used as an operational surrogate

**GA ≤34 weeks**
**Any respiratory distress**

**GA >34 weeks**
**Moderate–severe RD (SAS ≥4)**

**GA >34 weeks**
**Mild RD (SAS ≤3)**

### START CPAP

CPAP 5–6 cm $H_2O$; blended oxygen; titrate $FiO_2$ to $SpO_2$ 91–95%
Caffeine citrate for babies <34 weeks requiring respiratory support, as per protocol

### NASAL PRONGS OXYGEN

If $SpO_2$ <91%: 0.5–1 L/min
Titrate to $SpO_2$ 91–95%

### MONITORING

**Monitor $SpO_2$ continuously while on oxygen/CPAP**
Reassess clinical status, SAS and $FiO_2$ requirement within 15–30 min of starting/changing support
Reassess at least 1 hourly while on respiratory support; immediately if any worsening

### IMPROVING

- Decreasing SAS
- $SpO_2$ 91–95% on stable/reducing support
- Comfortable breathing

### WEAN WHEN STABLE

- Comfortable breathing; no significant apnoea; low oxygen requirement
- Reduce $FiO_2$, then CPAP as per protocol
- Trial off when stable on CPAP 5 cm $H_2O$ and $FiO_2$ 0.21–0.25
- Monitor $SpO_2$ for 24 h after stopping

### EARLY RESCUE SURFACTANT

- Suspected worsening RDS in a preterm neonate <34 weeks, AND
- **$FiO_2$ >0.30 on CPAP ≥6 cm $H_2O$** to maintain $SpO_2$ 91–95%

Give promptly when indicated—before established CPAP failure

### NOT IMPROVING / WORSENING

- Persisting/increasing SAS
- $SpO_2$ <91% or rising oxygen need
- Recurrent apnoea/bradycardia
- **Consider sepsis (Refer to sepsis STW)**

### OPTIMIZE RESPIRATORY SUPPORT

- If on Nasal Prongs oxygen, rising oxygen need or distress: Start CPAP
- Check prongs, circuit and delivered pressure
- Increase CPAP up to 7–8 cm $H_2O$; Titrate $FiO_2$
- Exclude pneumothorax (Transillumination, X-ray)
- Consider surfactant

### CPAP FAILURE — ESCALATE PROMPTLY

Persistent hypoxaemia/rising $FiO_2$ despite optimized CPAP, recurrent apnoea, shock or fatigue
**Escalate support; intubate/ventilate when indicated; stabilize and refer urgently**

### DOs

- Assess severity using SAS; monitor $SpO_2$ continuously
- Maintain $SpO_2$ 91–95%
- Start CPAP within 30 min of identification of indication
- Give early rescue surfactant promptly when indicated

### DON'Ts

- Do NOT give unmonitored supplemental oxygen
- Do NOT order routine CBC/CRP, chest X-ray or blood gas in uncomplicated mild RD
- Do NOT give IV fluids, antibiotics or blood products routinely
- Do NOT delay ventilation or referral when CPAP fails

### KEY PERFORMANCE INDICATORS (KPIs)

**Timely CPAP initiation (%)**

$$\frac{\text{Eligible neonates with RD started on CPAP within 30 min of identification}}{\text{All neonates with RD meeting the STW criteria for CPAP}} \times 100$$

**Target >90%**

**Continuous pulse-oximetry coverage (%)**

$$\frac{\text{Neonates receiving oxygen or CPAP with documented continuous pulse oximetry}}{\text{All neonates receiving oxygen or CPAP}} \times 100$$

**Target >90%**

**ASSESS SAS • PROVIDE APPROPRIATE RESPIRATORY SUPPORT • TARGET $SpO_2$ 91–95%**
**ESCALATE SUPPORT PROMPTLY IF NOT IMPROVING**

### RELATED INFORMATION

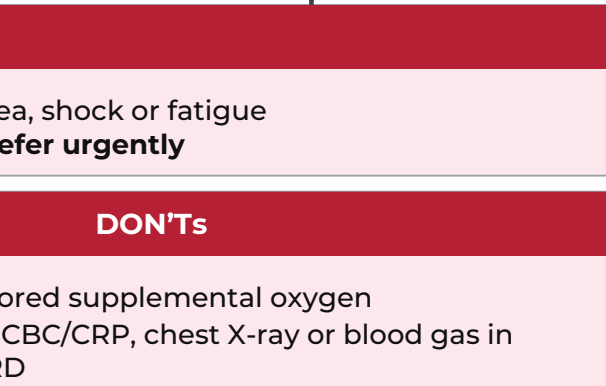
Bubble CPAP setup

Fixing binasal prongs

SAS scoring

Surfactant administration

### SILVERMAN–ANDERSEN SCORE




SAS: Avery & Fletcher (1974); W. A. Silverman

### ABBREVIATIONS

BW: Birth weight
CPAP: Continuous positive airway pressure
CRT: Capillary refill time
GA: Gestational age
$FiO_2$: Fraction of inspired oxygen
PEEP: Positive end-expiratory pressure
RD: Respiratory distress
RR: Respiratory rate
SAS: Silverman–Andersen score
$SpO_2$: Peripheral oxygen saturation
TABC: Temperature, airway, breathing, circulation

### REFERENCES

1. National Neonatology Forum India. Oxygen Therapy in Neonates; Surfactant Replacement Therapy in Neonates. Evidence-based Clinical Practice Guidelines. nnfi.org/cpg.php
2. Sweet DG et al. European Consensus Guidelines on the Management of Respiratory Distress Syndrome: 2022 Update. Neonatology. 2023;120:3–23.

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